

- Q.26 Write a short note on I/P converter.
- Q.27 Explain all pneumatic control elements.
- Q.28 What is control valves & explain butterfly valve.
- Q.29 Describe solenoid valve.
- Q.30 Explain the diaphragm operated valve.
- Q.31 Describe interlocking & sequencing circuit.
- Q.32 Differentiate between Temp switch & pressure switch.
- Q.33 Draw the basic structure of control valves is used in control system.
- Q.34 Describe flapper nozzle system in control system.
- Q.35 Describe proportional controller in a control system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain about PID controller. Also write the advantages of PID controller.
- Q.37 Attempt any two:
- Auto transformer
 - I/P converter
 - Electric actuator
- Q.38 Explain the process with example of all process variables (i.e. Liquid Level Controller)

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the standard current value in I/P converter:-
- 4-20 mA
 - 0-20 mA
 - 4-16 mA
 - 0-16 mA
- Q.2 Actuators are works as:-
- Final Control Elements
 - Relay
 - Pneumatic Actuator
 - Hydraulic actuator
- Q.3 Derivative controller is also known as.
- Rate controller
 - Reset controller
 - Both of these
 - None of these
- Q.4 Which of the following is the unit of sound:-
- Kelvin
 - Pascal
 - Newton
 - Decibal

- Q.5 Rotameter fixed in:-
 a) Vertical Position b) Horizontal Position
 c) Any position d) Tapered Position
- Q.6 “Accuracy” defined as-
 a) Closeness to the true value
 b) Repeatability
 c) Never changing value
 d) All of true
- Q.7 For fast response which control required
 a) P-Controller b) I- Controller
 c) D- Controller d) All of above
- Q.8 Which of the following is the dynamic characteristics of an instrument-
 a) Sensitivity b) Reproductively
 c) Fidelity d) Dead Zone
- Q.9 PD stands for:-
 a) Propositional Derived
 b) Propositional Derivative
 c) Positional Derived
 d) Positional Derivative
- Q.10 Process Lag means:-
 a) Process is advanced b) Process is delayed
 c) Process has no effect d) None of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is transmission lag.
 Q.12 SI Unit of Pressure.
 Q.13 Write down 2 merits of Integral controller.
 Q.14 What is an amplifier.
 Q.15 What is hydraulic Actuator.
 Q.16 Write down 3 types of control valve.
 Q.17 What are the process variables.
 Q.18 Define resolution.
 Q.19 Expand SCADA.
 Q.20 What is process lag.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain all process variables, those are considered in a process.
 Q.22 Describe process lag & measurement lag.
 Q.23 Write a short note on ON-OFF control mode.
 Q.24 Describe PI controller & its merits also.
 Q.25 Differentiate between Pneumatics & hydraulic systems.

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